

Lesson 4



The compensated law of demand iff statement

so the two parts of the statement are equivalent. Then the same thing.

so WARP is the same as

$$(P' - P)(x(P', u') - x(P, u)) \leq 0 \text{ when } w' = P' \cdot x(P, w)$$

We prove WARP $\rightarrow$ inequality

$$\text{If } x(P', u') = x(P, u) \text{ since } x(P', u') - x(P, u) = 0 \\ \text{assume } x(P', u') \neq x(P, u)$$

From the assumption

$$P' \cdot x(P, w) = w' = P' \cdot x(P', w')$$

$$P'(x(P', u') - x(P, w))$$

$$\text{Hence } P'(x(P', u') - x(P, u)) \geq 0$$

Time to use WARP + $x(P', u') \neq x(P, w)$

$\Rightarrow x(P', w')$ is R.P. to $x(P, u)$, so WARP tells

us that $x(P, w)$ was picked under (P, w) because $x(P', w')$ was not affordable

$$P \cdot x(P', w') > w \quad \text{WARP,}$$

$$> P \cdot x(P, w)$$

$$P(x(P', w') - x(P, w)) > 0$$

$$(P' - P)(x(P', u') - x(P, w)) < 0$$

We move to key results regarding marginal price change
 Consider change in x_i when p_i change marginally at
 the same due to wealth compensation.

Wealth changes to $w' = p' \cdot x(p, w)$

hence demand is $x_i(p', p'_i x(p, w))$

Assuming differentiability

$$\frac{\partial x_i(p', p' \cdot x(p, w))}{\partial p'_j} = \frac{\partial x_i(p', p' \cdot x(p, w))}{\partial p_j}$$

$$+ \frac{\partial x_i(p', p' \cdot x(p, w))}{\partial w} x_j(p, w)$$

$$\left. \frac{\partial x_i(p', p' \cdot x(p, w))}{\partial p_j} \right|_{p'=p} = \frac{\partial x_i(p, w)}{\partial p_j} + \frac{\partial x_i(p, w)}{\partial w} x_j(p, w)$$

The substitution effect S_{ij} on Good i when price of good j changes

The $L \times L$ matrix that collects all S_{ij} 's is the substitution or Slutsky matrix.

What can we say about the ~~stri~~ structure of the substitution matrix?

A clue is the compensated law of demand

in $\sum_{i=1}^L dP_i dx_i \leq 0$

the change in wealth is $dw = x(P, u) dP$
 $= \sum_{i=1}^L x_i(P, w) dP_i$

$$dx_i = \sum_{j=1}^L \frac{\partial x_i(P, w)}{\partial P_j} dP_j + \frac{\partial x_i(P, w)}{\partial w} dw$$

$$= \sum_{j=1}^L \frac{\partial x_i(P, w)}{\partial P_j} dP_j + \frac{\partial x_i(P, w)}{\partial w} \sum_{i=1}^L x_i(P, w) dP_j$$

$$= \sum_{i=1}^L \frac{\partial x_i(P, w)}{\partial P_j} dP_j + \frac{\partial x_i(P, w)}{\partial w} x_i(P, w) dP_j$$

$$= \sum_{i=1}^L S_{ij} dP_j$$

We therefore have

$$\sum_{i=1}^L dP_i dx_i = \sum_{i=1}^L dP_i \sum_{j=1}^L S_{ij} dP_j$$

$$= \sum_i \sum_j dP_i S_{ij} dP_j \leq 0 \text{ by the compensated law of the demand.}$$

$\Rightarrow$ The Slutsky matrix is negative semi-definite.

Later, we get the same conclusion when we impose axioms on preferences.

An example that nevertheless suggest that WARP on its own is not enough to establish the agent has rational preference reactions

$p^0 = 1, 1, 2$	$p^1 (1, 1, 1)$	$p^2 (1, 2, 1)$
$x^0 (5, 19, 9) \quad 42$	33	52
$x^1 = (12, 12, 12) \quad 48$	36	48
$x^2 = (27, 11, 1) \quad 40$	39	<u>50</u>
revealed	x^0 R.P. x^2	x^1 R.P. x^0
	x^0 not affordable at $p^2 = \text{WARP}$	x^2 R.P. to x^1
	✓	✓

$\Rightarrow$ x^0 R.P. x^2 R.P. x^1 R.P. x^0
Intransitive

WARP is not enough

Strong axiom A.P. (SARP) Assume ^{away} transitivity cycle.